Halvorsen Bioprocess AB

Kvarnvagen 12, 752 28 Uppsala, Sweden

Material Description Sheet: FlowLine Single-Use Kits

Document No.: MDS-FL-2024-02

Materials of Construction

Component	Material
Tubing	Platinum-cured silicone
Connectors	Polycarbonate
Mixing chamber	Polypropylene
Pressure sensor housing	Polysulfone

Physical Properties

Property	Value
Operating temperature	+4 C to +37 C
Maximum operating pressure	3.0 bar
Kit weight	0.85 kg

Sterilization Compatibility

Supplied gamma-irradiated. Not suitable for autoclaving.

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Certificate of Quality

Document No.: HQ-2024-1183

Field	Value
Product	FlowLine Low Flow Kit
Lot Number	HB-77310
Certificate Date	9 October 2024
Expiry Date	36 months from manufacture

Test Results

Test	Specification	Result
Sterility	Sterile	Pass
Endotoxin	< 0.25 EU/mL	< 0.05 EU/mL
Leak test	No leak at 3.0 bar	Pass

Approved by:

Elin Bergqvist, QA Release Officer

Date: 9 October 2024

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Certificate of Quality

Document No.: HQ-2024-1207

Field	Value
Product	FlowLine High Flow Kit
Lot Number	HB-77452
Certificate Date	16 October 2024
Expiry Date	36 months from manufacture

Test Results

Test	Specification	Result
Sterility	Sterile	Pass
Endotoxin	< 0.25 EU/mL	< 0.05 EU/mL
Leak test	No leak at 3.0 bar	Pass

Approved by:

Elin Bergqvist, QA Release Officer

Date: 16 October 2024

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Packaging Component Specification

Document No.: PCS-FL-011

Scope

Packaging configuration for FlowLine Single-Use Kits.

Primary Packaging

Component	Material / Grade
Blister tray	PETG, 0.8 mm
Lid	Medical-grade Tyvek

Continued on next page.

Packaging Component Specification - continued (PCS-FL-011)

Secondary Packaging

Component	Material / Grade
Outer pouch	Polyethylene / nylon laminate
Shipping box	Corrugated cardboard

Revision History

Rev	Date	Change
A	Mar 2022	Initial release
B	Jul 2024	Lid material changed to Tyvek

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BSE/TSE Declaration

Document No.: BSE-FL-2024

Halvorsen Bioprocess AB declares that FlowLine Single-Use Kits contain no materials of animal origin (bovine, ovine, caprine, or porcine).

This declaration is based on a BSE/TSE risk assessment in accordance with EMA/410/01 rev. 3. Assessment reference: TSE-RA-2024-FL-003.

Approved by:

Elin Bergqvist, QA Release Officer

Date: 2 September 2024

Halvorsen Bioprocess AB

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Global Chain of Custody

Document No.: COC-FL-2024

Prepared by: Supply Chain Quality, Halvorsen Bioprocess AB

Assemblies Covered

FlowLine Low Flow Kit; FlowLine High Flow Kit

Supply Chain Steps

1. Raw material receipt and incoming inspection (Uppsala, Sweden)
2. Component molding and tubing extrusion (Uppsala, Sweden)
3. Cleanroom assembly and packaging (Tallinn, Estonia)
4. Gamma sterilization (contract sterilizer, Germany)
5. Distribution to end customer

Traceability Statement

Each kit carries a unique lot number traceable to raw material lots, assembly date, and cleanroom operator records.